    - Perform the semantic search, and compare the results with those obtained using TF–IDF to understand the differences and effectiveness.
- *Exercise 3* —Implement a Custom RAG Workflow (Advanced)

Objective —Apply theoretical knowledge of RAG in a practical context using LangChain.

Tasks:

    - Choose a specific application (e.g., customer service inquiries or academic research queries).
    - Design and implement a custom RAG workflow using LangChain.
    - Tailor the workflow to suit the chosen application, and test its effectiveness.
- *Exercise 4* —Build a Knowledge Store and Experiment with Splitting Patterns (Intermediate)

Objective —Understand how different splitting patterns and compression affect knowledge retrieval.

Tasks:

    - Build a knowledge store, and populate it with a couple of documents.
    - Experiment with different forms of splitting/chunking patterns, and analyze their effect on retrieval.
    - Compress the knowledge store, and observe the effects on query performance.
- *Exercise 5* —Build and Test Various Memory Stores (Advanced)

Objective —Understand the uniqueness and use cases of different memory store types.

Tasks:

    - Build various forms of memory stores (conversational, semantic, episodic, and procedural).